

Nylander's Test

As a reducing sugar, glucose reacts in the Nylander's test.^[110]

Other tests

Upon heating a dilute potassium hydroxide solution with glucose to 100 °C, a strong reddish browning and a caramel-like odor develops.^[111] Concentrated sulfuric acid dissolves dry glucose without blackening at room temperature forming sugar sulfuric acid.^[111] In a yeast solution, alcoholic fermentation produces carbon dioxide in the ratio of 2.0454 molecules of glucose to one molecule of CO₂.^[111] Glucose forms a black mass with stannous chloride.^[111] In an ammoniacal silver solution, glucose (as well as lactose and dextrin) leads to the deposition of silver. In an ammoniacal lead acetate solution, white lead glycoside is formed in the presence of glucose, which becomes less soluble on cooking and turns brown.^[111] In an ammoniacal copper solution, yellow copper oxide hydrate is formed with glucose at room temperature, while red copper oxide is formed during boiling (same with dextrin, except for with an ammoniacal copper acetate solution).^[111] With Hager's reagent, glucose forms mercury oxide during boiling.^[111] An alkaline bismuth solution is used to precipitate elemental, black-brown bismuth with glucose.^[111] Glucose boiled in an ammonium molybdate solution turns the solution blue. A solution with indigo carmine and sodium carbonate destains when boiled with glucose.^[111]

Instrumental Quantification

Refractometry and Polarimetry

In concentrated solutions of glucose with a low proportion of other carbohydrates, its concentration can be determined with a polarimeter. For sugar mixtures, the concentration can be determined with a refractometer, for example in the Oechsle determination in the course of the production of wine.

Photometric enzymatic methods in solution

The enzyme glucose oxidase (GOx) converts glucose into gluconic acid and hydrogen peroxide while consuming oxygen. Another enzyme, peroxidase, catalyzes a chromogenic reaction (Trinder reaction)^[112] of phenol with 4-aminoantipyrine to a purple dye.

Photometric Test Strip Method

The test strip method employs the above-mentioned enzymatic conversion of glucose to gluconic acid to form hydrogen peroxide. The reagents are immobilised on a polymer matrix, the so-called test strip, which assumes a more or less intense color. This can be measured reflectometrically at 510 nm with the aid of an LED-based handheld photometer. This allows for routine blood sugar determination by laymen. In addition to the reaction of phenol with 4-aminoantipyrine, new chromogenic reactions have been developed that allow photometry at higher wavelengths (550 nm, 750 nm).^[113]

Amperometric glucose sensor

The electroanalysis of glucose is also based on the enzymatic reaction mentioned above. The produced hydrogen peroxide can be amperometrically quantified by anodic oxidation at a potential of 600 mV.^[114] The GOx is immobilised on the electrode surface or in a membrane placed close to the electrode. Precious metals such as platinum or gold are used in electrodes, as well as carbon nanotube electrodes, which e.g. are doped with boron.^[115] Cu-CuO nanowires are also used as enzyme-free amperometric electrodes. This way a detection limit of 50 µmol/L has been achieved.^[116] A particularly promising method is the so-called "enzyme wiring". In this case, the electron flowing during the oxidation is transferred directly from the enzyme via a molecular wire to the electrode.^[117]

Other sensory methods

There are a variety of other chemical sensors for measuring glucose.^{[118][119]} Given the importance of glucose analysis in the life sciences, numerous optical probes have also been developed for saccharides based on the use of boronic acids,^[120] which are particularly useful for intracellular sensory applications where other (optical) methods are not or only conditionally usable. In addition to the organic boronic acid derivatives, which often bind highly specifically to the 1,2-diol groups of sugars, there are also other probe concepts classified by functional mechanisms which use selective glucose-binding proteins (e.g. concanavalin A) as a receptor. Furthermore, methods were developed which indirectly detect the glucose concentration via the concentration of metabolised products, e.g. by the consumption of oxygen using fluorescence-optical sensors.^[121] Finally, there are enzyme-based concepts that use the intrinsic absorbance or fluorescence of (fluorescence-labeled) enzymes as reporters.^[118]

Copper iodometry

Glucose can be quantified by copper iodometry.^[122]

Chromatographic methods

In particular, for the analysis of complex mixtures containing glucose, e.g. in honey, chromatographic methods such as high performance liquid chromatography and gas chromatography^[122] are often used in combination with mass spectrometry.^{[123][124]} Taking into account the isotope ratios, it is also possible to reliably detect honey adulteration by added sugars with these methods.^[125] Derivatisation using silylation reagents is commonly used.^[126] Also, the proportions of di- and trisaccharides can be quantified.

In vivo analysis

Glucose uptake in cells of organisms is measured with 2-deoxy-D-glucose or fluorodeoxyglucose.^[76] (¹⁸F)fluorodeoxyglucose is used as a tracer in positron emission tomography in oncology and neurology,^[127] where it is by far the most commonly used diagnostic agent.^[128]